

TabXEval: Why this is a Bad Table?

An eXhaustive Rubric for Table Evaluation



*Vihang Pancholi¹, *Jainit Bafna², *Tejas Anvekar¹,
Manish Shrivastava², Vivek Gupta¹
¹ Arizona State University, ² IIIT Hyderabad



INTERNATIONAL INSTITUTE OF
INFORMATION TECHNOLOGY
HYDERABAD

Motivation

Film	Pre-nomination (before Jan. 14)	Post-nomination (Jan. 14 – Feb. 28)	Post-awards (after Feb. 28)	Total
The Martian	\$226.6 million	\$1.8 million	\$53.548	\$228.4 million
The Revenant	\$54.1 million	\$116.5 million	\$11.9 million	\$182.6 million
Mad Max: Fury Road	\$153.6 million	\$1.4 million	\$49.549	\$153.6 million
Bridge of Spies	\$70.8 million	\$23.9 million	\$1.7 million	\$72.3 million
The Big Short	\$44.6 million	\$10.3 million	\$5.5 million	\$44.6 million
Spotlight	\$28.8 million	\$13.7 million	\$1.6 million	\$38.1 million
Brooklyn	\$22.8 million	\$8.2 million	\$1.2 million	\$14.7 million
Room	\$5.2 million	\$5.2 mil	\$8.2 mil	\$14.7 mil

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Metric	chrF	h.score	bert.score	p.score	EM	Bleurt	TabEval
Value	52	86	81	75	3	-0.04	96

Existing metrics overlook structure, tolerate missing columns, mis-score numbers, brittle to formatting, opaque results.

chrF: Character overlap ignores missing columns; still scores high despite severe data loss.

H-Score: Header comparison overlooks alignment; grants credit although corresponding cells are completely empty.

BERTScore: Embedding similarity forgives column drop; tokens similar, numbers unchanged, score remains inflated.

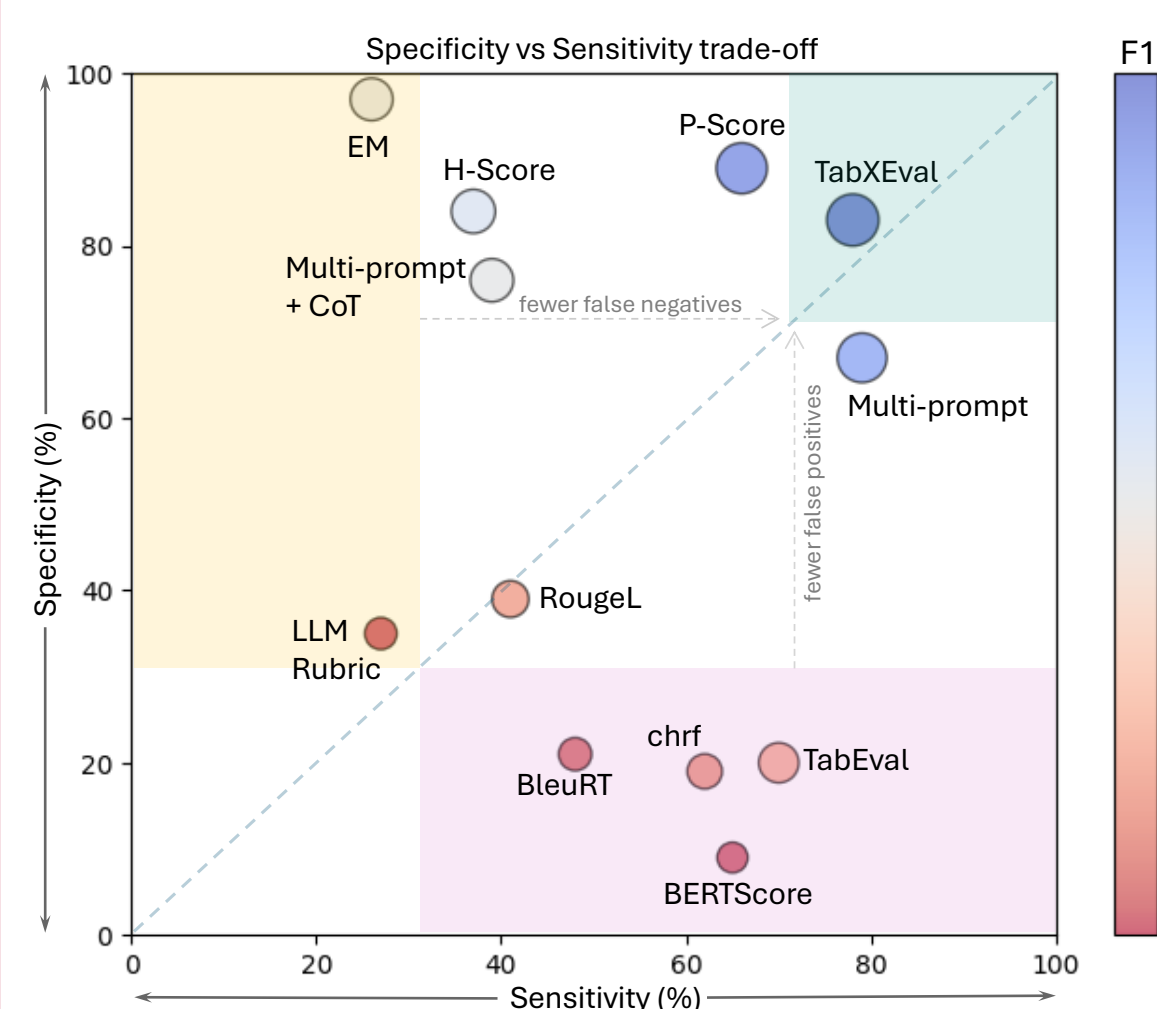
P-Score: LLM entailment detects issues vaguely; expensive, stochastic, no localisation of errors reported.

Exact Match: Exact match too brittle; formatting variations obliterate score despite factual correctness completely.

Bleurt: Sentence-level semantics misjudge tables; penalises formatting, misses column-scale numeric omissions entirely often.

TabEval: Heuristic metric inherits BERTScore blindness; applauds output despite missing Post-awards column completely.

Our Contributions

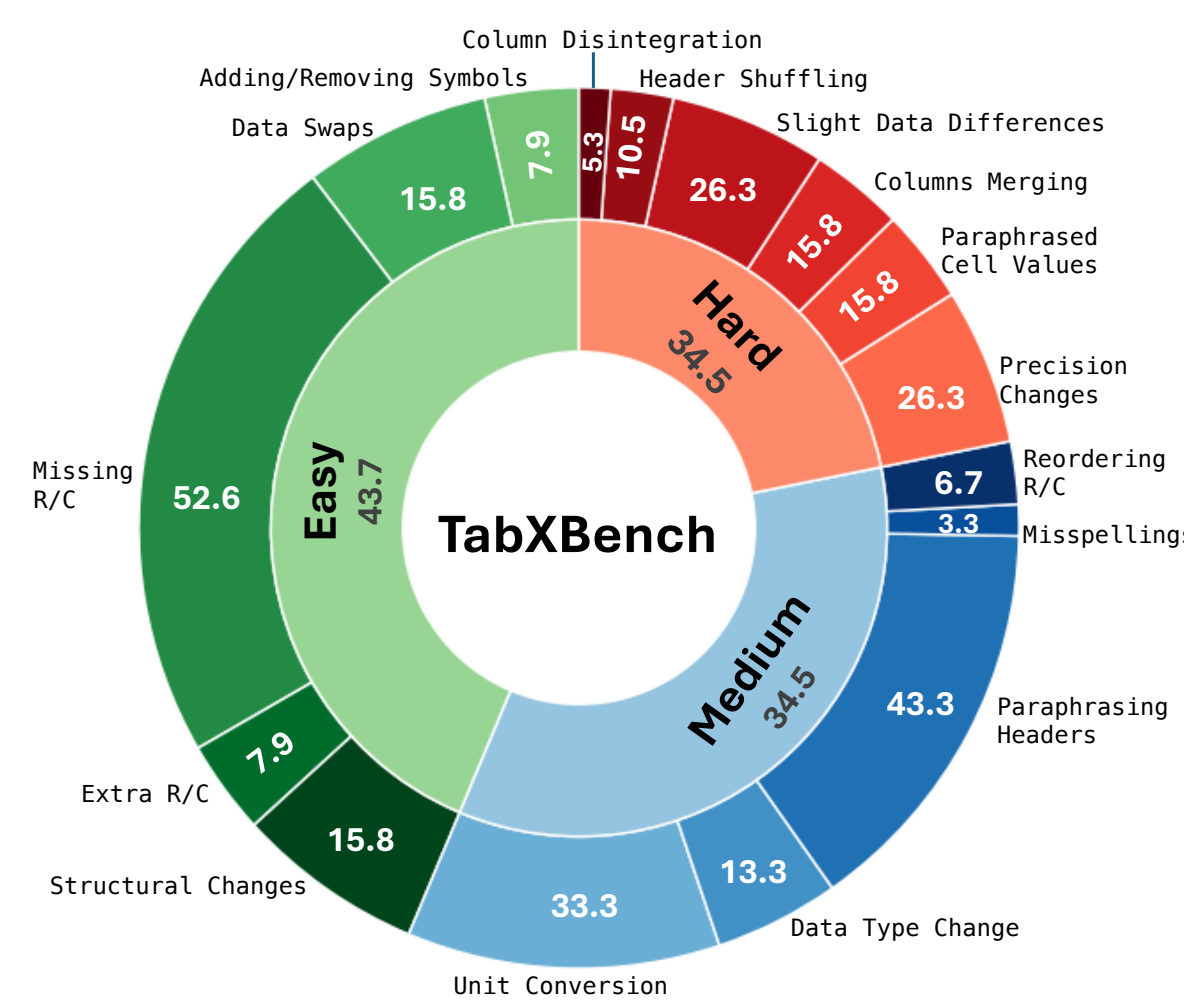


Gap in Current Evaluation Methods

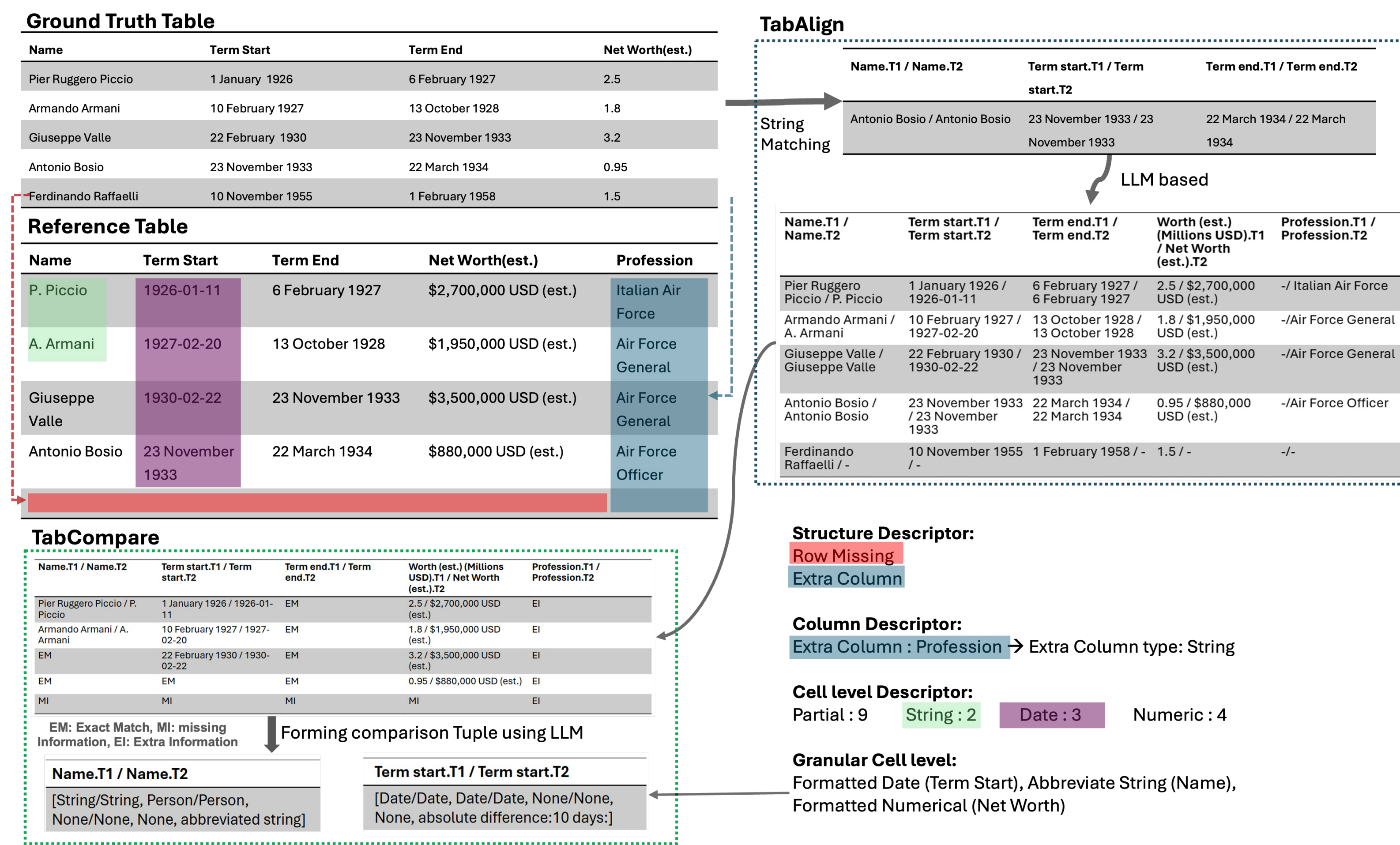
- Text-only view**: BLEU/ROUGE ignore table layout
- Semantics-only**: BERTScore misses row/col swaps
- Black-box scores**: P-/H-Score give no error trace
- Entailment heavy**: TabEval, numeric-blind because of use of BertScore

To Assess Table Evaluation Metric We Propose
TabXBench:

- 250 test cases
- 6 diverse sources
- 16 distinct error types
- 3 difficulty tiers
- Balanced stress test



TabXEval Overview



- 1st Rubric** that captures errors from **table structure** down to **individual cells**.

- Two-stage pipeline**: *TabAlign* aligns rows/columns; *TabCompare* labels cells as Exact, Missing, Extra, or Partial.

- Outputs a structured error matrix** for precise, interpretable diagnostics.

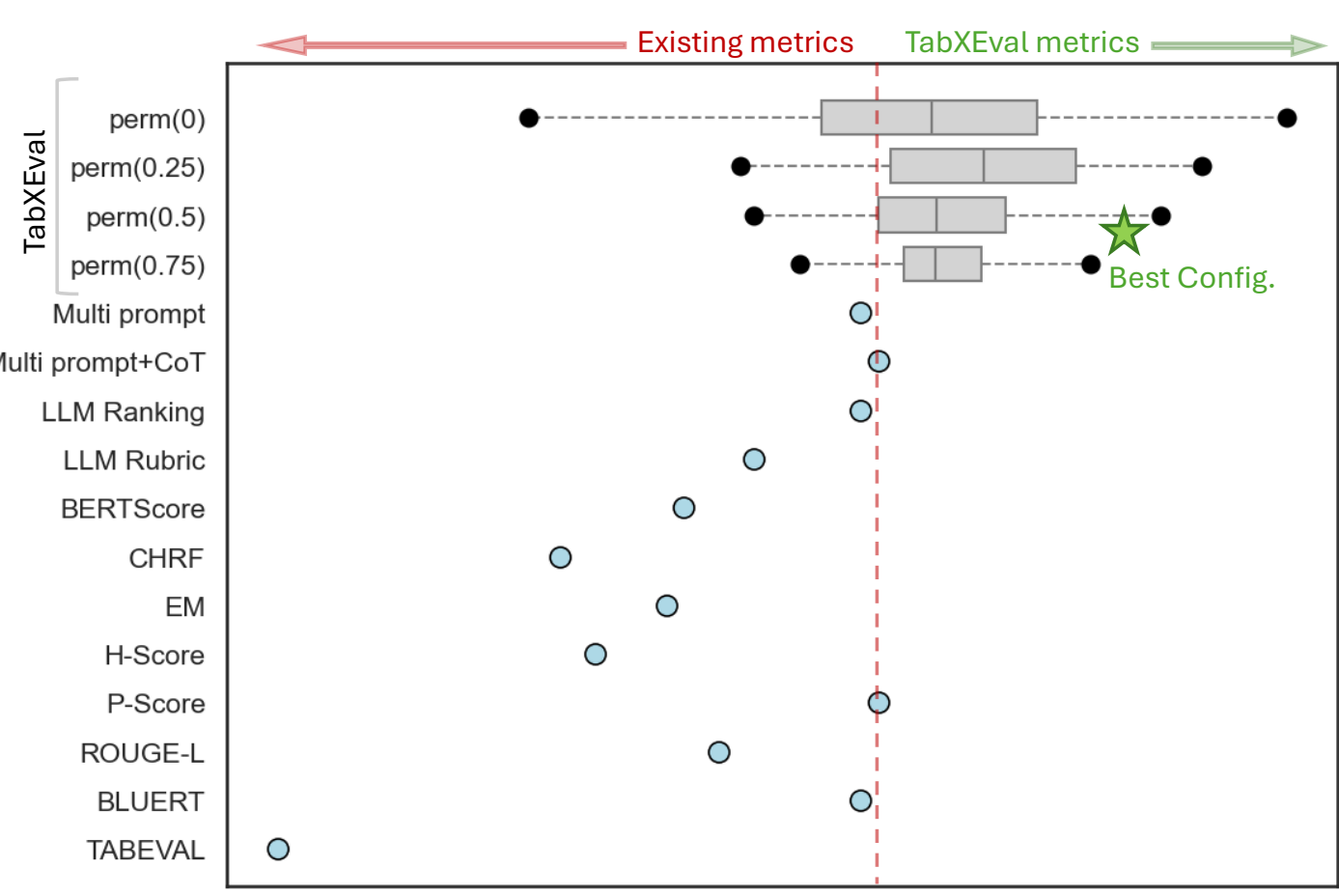
- Weight-based scoring** supports both domain-agnostic use and domain-specific tuning.

Analysis

LLaMA-3.3 70B									
Stat	Num	String	Bool	Date	List	Time	Others	Num	String
EM	0.05	4.33	0.00	0.17	0.00	0.13	0.03	0.00	0.00
MI	0.01	0.80	0.00	0.03	0.00	0.01	0.01	0.00	0.00
Partial	0.22	25.00	0.00	0.35	0.00	0.01	0.09	0.32	20.34

Table 3: **Cell-Level Performance Analysis** of Extra (EI), Missing (MI), and Partial mismatches across data types numerical, string, boolean, date, list, time, and other for WikiTables, WikiBio, TANQ, and RotoWire. **Highlights**: GPT-4o shows fewer string EI in WikiTables and lower partial errors in numerical and string cells in TANQ and RotoWire.

- Structure-level audit**: TabXEval aligns rows and columns to detect missing or extra structure, enabling interpretable comparison.
- Cell-level diagnostics**: Uses type-aware comparison to capture subtle semantic errors like unit mismatch and header drift.
- Model evaluation insights**: GPT-4o shows highest table fidelity (Row EM: 27%, Col EI: 0.07%) with minimal content error.
- Failure characterization**: LLaMA overgenerates noisy content; Gemini handles simple tables but struggles with complex ones.



- Domain-agnostic core** – outperforms all baselines on every corpus without tuning
- Domain-adaptive knobs** – re-weight structure vs semantics (row/col vs cell) for task focus.
- Balanced ★ config** – highest human-correlation, lowest variance → fidelity + flexibility.

Results

Method	Avg %	Spearman's ρ	Kendal's τ	RBO
TabXEval	39.3%	44%	40%	34%
P-Score	29.3%	30%	27%	31%
MP + CoT	28.0%	30%	25%	29%
LLM Rank	27.6%	29%	24%	30%
MP	27.6%	29%	24%	30%
BLUERT	27.0%	29%	25%	27%
LLM Rubric	22.3%	23%	16%	28%
ROUGE-L	22.0%	21%	18%	27%
EM	20.0%	18%	16%	26%
BERTScore	19.6%	19%	15%	25%
H-Score	17.6%	14%	11%	28%
CHRF	16.0%	12%	11%	25%
TabEval	5.0%	10%	10%	23%

- TabXEval +15-20 pp** over LLM-prompted metrics → best structure + semantics
- P-Score / BLEURT**: Plateau despite improvements over lexical baselines.
- TabEval lowest** – many false positives on numeric / unit errors

Take-Aways

- Introduced **TabXEval**, a rubric-driven, two-phase framework and **TabXBench**, a broad benchmark for reliable table evaluation.

- Exposed major blind spots in current metrics; TabXEval delivers the best human-aligned scores while remaining domain-agnostic and explainable.

Try TabXEval Yourself at:

<https://coral-lab-asu.github.io/tabxeval/>

